

Curriculum Vitae

Dr. Anurag Tripathi

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EMPLOYMENT/EDUCATION

Associate Professor	IIT Hyderabad, India	14.10 2021 - till date
Assistant Professor	IIT Hyderabad, India	20.07 2015 - 13.10. 2021
Postdoc. Fellow	INFN Naples, Italy	03.11. 2014 - 14.07. 2015
Postdoc. Fellow	Univ. of Turin & INFN Turin, Italy	01.11. 2013 - 31.10. 2014
Postdoc. Fellow	Univ. of Wuppertal, Germany	14.10. 2011 - 15.10. 2013
Postdoc. Fellow	Tata Inst. of Fundamental Res., India	20.10. 2009 - 23.09. 2011
Ph.D+Course Work	Harish-Chandra Res. Inst., India	02.08. 2003 - 18.08. 2009
Thesis Title	Higher order QCD radiative corrections to processes at the Large Hadron Collider.	
Thesis Advisor	Prof. V. Ravindran.	
Submission Date	18 Aug. 2009.	
Defense Date	02 Feb. 2010.	
Award Date	02 Feb. 2010.	

RESEARCH INTERESTS

- QCD corrections to strong interaction processes.
- Infrared structure of scattering amplitudes, soft anomalous dimension.
- Resummation of soft and collinear gluons.
- Higgs phenomenology
- Power corrections to event shapes at e^+e^- and hadron colliders.

SCHOLARSHIPS/AWARDS/Academic Recognition

1. Joined as a member of the Editorial Board of Journal of Physics G, IOP Publishing in Jan 2025.
2. Grant awarded by MHRD under SPARC scheme. Title: *Perturbative QCD for Precision Physics at the LHC*. Grant amount. INR Eighty One Lacs (81 Lacs). Duration: 15.03.2019 - 30.09.2023. Project code P578.
3. Grant awarded by MHRD under GIAN scheme for conducting a course on *Infrared Structure of Perturbative Gauge Theories*. Grant amount INR Eight Lacs. Award year 2017. Project Code: 171008M01. GIAN course duration: 04 July -13 July 2018.
4. Qualified National Eligibility Test for Lecturership (NET), 2003, India.
5. Qualified CSIR-UGC Junior Research Fellowship (JRF), 2003, India.
6. Qualified GATE (Graduate Aptitude Test in Engineering) with 96.40 Percentile, 2003, India.

SWAYAM/ NPTEL COURSES (MOOC) OFFERED

- Introduction to Classical Mechanics. 2020, 2021, 2022, 2023 [Click here for the Link](#)
- Introduction to Quantum Field Theory (Theory of Scalar Fields) Part I. 2021,2022, 2023 [Click here for the Link](#)
- Introduction to Quantum Field Theory (Theory of Scalar Fields) Part II. 2023, 2024 [Click here for the Link](#).

OUTREACH ACTIVITIES

- Conducted a hands-on session for 100 Navodaya Vidyalaya Teachers (Nov12-13 & 27-28, 2024).
- 8-bit computer making session for Research exposure camp for Physics College Teachers team of Prof. H.C. Verma (26 Feb 2025).

ACADEMIC ACTIVITIES

- Formed an informal club – *Builders' Society of IIT Hyderabad Junior Wing* – in April 2025. This club has around 300 members who build things and help each other. Students have build 8-bit computers, Van de Graff generators etc.
- Organized International Workshop "International Workshop: Precision QCD @LHC" at IIT Hyderabad in Jan 2020.
- Organized GIAN program at IITH in 2017.
- Guided undergraduate students visiting me for summer internship.

ADMINISTRATIVE ACTIVITIES

- Convenor of Green Office, from 30 September 2019 - 30 August 2022.
- Nominated as a member from Physics Department for Vision 2030 Committee (2022)
- Coordinator, Biodiversity Committee, from 24 April 2016 - 30 September 2019
- Coordinator for PhD admissions in 2016.
- Coordinator for Physics Department Brochure development at IITH in 2016.

AWARDS TO MY PHD STUDENTS

- Sourav Pal and Abhinava Danish received Delta-ITP fellowship to visit Nikhef and the University of Amsterdam for collaboration with Prof. Eric Laenen's group in 2019.
- Sourav Pal also received support from INFN Turin and the University of Turin to visit the group of Prof. Lorenzo Magnea at the department of Physics for a period of 5 weeks in 2019.

SUPERVISING EXPERIENCE

- Postdoctoral Fellows (02)
 1. Dr. Tanmay Maji (2022) (NPDPF)
 2. Dr. Debashis Saha (2024) (NPDPF)
- Ph.D Students Graduated (04) (Graduation Year)
 1. Ayan Mukhopadhyay (2023)
 2. Sourav Pal (2023)
 3. Shubham Mishra (2025)
 4. Aditya Srivastav (2025)
- Ph.D Students (05) (Joining Year)
 1. Debojyoti Banerjee (2024)
 2. Abhinava Danish (2018)
- M.Sc. Students (03)

Ashish Chandra (2016-2017), Abhinandan Yadav (2017-2018), Laboni Manna (2018-2019), Praveen Patil (2020-2021), Trina Basu (2023), Ashish Arun Kurdekar (2024), Sajid Ali (2024)
- Enigneering Physics Students (03)

Mamidi Ratna Lahari (2019), Raad Shaikh (2020), Utkarsh Bhura (2020), Amaan (), Nanjundaradhya H Y (2024)

SUMMARY OF IMPORTANT CONTRIBUTIONS

- **Precision BSM physics:** During my Ph.D., I focused my work on precision studies of observables at the Hadron Colliders. These studies were mainly aimed at beyond the Standard Model Physics. I calculated cross-sections for producing a di-photon final state at the LHC at next-to-leading order in QCD. Using my works, the ATLAS and the CMS collaborations at the LHC have put stronger constraints on the extra dimension models (CERN-PH-EP-2012-279, CERN-PH-EP-2012-289, CMS-PAS-EXO-11-038). My publications — NPB 818 (2009), PLB 672 (2009), have been cited over 150 times in the literature.
- **Resummed Higgs production cross-sections:** Fixed order results for transverse momentum distributions diverge as $\mathbf{p}_T \rightarrow 0$ due to the soft and collinear enhancements. But small $\mathbf{p}_T$ region is important because cross-sections are large in this region and a significantly larger amount of data from experiments is available in this region compared to large $\mathbf{p}_T$. The predictive power of the theory can be restored if these contributions are resummed to all orders in perturbation theory. I worked in this direction in two of my publications:
 - In Phys. Rev. D 90, 015017 we presented the inclusive transverse momentum distribution for Higgs bosons produced in bottom quark annihilation at the LHC. The soft and collinear terms at small $\mathbf{p}_T$ are resummed through NNLL accuracy and matched to the NNLO transverse momentum distribution at large $\mathbf{p}_T$.
 - In JHEP 12 (2018) we have studied the transverse momentum distribution of the pseudo-scalar Higgs boson at the Large Hadron Collider (LHC). Using the universal infrared behaviour of the QCD we resum these large logarithms up to next-to-next-to-leading logarithmic (NNLL) accuracy.
- **Event shapes:** These are classical tools for the determination of the strong coupling and for the study of hadronization effects in electron-positron annihilation.
 - In the context of analytical studies, hadronization corrections take the form of power-suppressed contributions to the cross section, which can be extracted from the perturbative ambiguity of Borel-resummed distributions. We have recently propose a simplified version of the well-established method of Dressed Gluon Exponentiation (DGE), which we call Eikonal DGE (EDGE), which determines all dominant power corrections to event shapes by means of strikingly elementary calculations. We believe our method can be generalized to hadronic event shapes and jet shapes of relevance for LHC physics. My method has been published in JHEP 03 (2021) 155.
 - In another work on event shapes (Phys.Rev.D 109 (2024) 9, 094038) we investigate the origin of next-to-leading power corrections to the event shapes thrust and c-parameter, at next-to-leading order. For both event shapes we trace the

origin of such terms in the exact calculation, and compare with a recent approach involving the eikonal approximation and momentum shifts that follow from the Low-Burnett-Kroll-Del Duca theorem.

- **Infrared structure of multiparton scattering amplitudes:** Understanding the IR structure of multiparton scattering amplitudes at 4-loops is one of the frontiers of research in High Energy Physics. It is a theorem that the scattering amplitudes exponentiate in the infrared region - the region where the virtual particles in the loops become soft or collinear to the external partons. The object of interest is the soft anomalous dimension that sits in the exponent. Several groups worldwide are actively pursuing computations of parts that would make the story at 4-loops complete and slowly — because the computations are challenging, but steadily, the community is moving forward.

There are two approaches; one is based on the renormalisation group approach, and the alternative is that of *Webs*, which is diagrammatic in nature. Efforts are underway to compute the webs at 4-loops. There are two ingredients to this: the kinematic factors of each of the webs and the corresponding colour factors.

- Since 2018, this latter aspect has been the focus of our research. Since then, we have developed from scratch a Wolfram Language (Mathematica) code for symbolic computation, which runs into thousands of lines. This is the only code available that calculates the web mixing matrices and exponentiated colour factors that enter the exponent at 4 and 5 loops.
- In 2020, and 2021 we published the results for all the multi-parton amplitudes that appear at 4 loops (JHEP 05 (2020) 128, JHEP 05 (2020) 128). Furthermore, we made an interesting discovery that for certain webs the constraints coming from row sum rule and column sum rule together with the idempotence of the the mixing matrices are sufficient to fully constrain the form of the mixing matrices to all orders in perturbation theory.
- In 2022 we made major advances in understanding the building blocks of the Web mixing matrices. In (JHEP 06 (2022) 020)) we introduce several new concepts: (a) Normal ordering of the diagrams of a Cweb, (b) Fused-Webs (c) Basis and Family of Cwebs. We use these ideas together with a Uniqueness theorem that we prove to arrive at an understanding of the diagonal blocks, and several null matrices that appear in the mixing matrices. We demonstrate using our formalism that, once the basis Cwebs present upto order α_s^n are determined, the number of exponentiated colour factors for several classes of Cwebs starting at order α_s^{n+1} can be predicted.
- In the beginning of 2023, we used our newly developed formalism (JHEP 02 (2023) 258) to Decipher colour building blocks of massive multiparton amplitudes at 4-loops and beyond; and in 2307.15924 [hep-ph] we gave an improved version of the algorithm for finding Cwebs and obtained the entire Cweb mixing matrices at five loops.

- In 2024 and 2025 we published three papers which provide explicit mixing matrices for Cwebs of massive multi-parton amplitudes (Phys. Rev. D 109 (2024) 094038), Cwebs at 5 loops (JHEP 07 (2024) 078 [2305.17452]) and the colour structure for next-to-eikonal Cwebs at three loops (Phys. Rev. D 111 (2025) 114022).
- The referees of our papers have been requesting to make our code CwebGen public which we have finally done. This code is very user friendly and available on Github at <https://github.com/tripathianurag80/CWebGen>. This work has been published in Computer Physics Communications 316 (2025), 109765

This series of work on Cwebs provided mixing matrices and exponentiated colour factors that enter the expression of the Soft function for multi-parton scattering amplitudes at three loops and beyond. Also the code CWebGen, which has now been made public, will definitely accelerate further developments in the study of Soft Anomalous dimension of multi-parton scattering amplitudes. The calculation of kinematical contributions at 4-loops prove to be difficult to obtain, though some partial progress has been made by some groups, which is crucial to complete the picture at 4-loops. Nevertheless, since the colour and kinematic contributions enter the soft function as a matrix product, it is believed that the results obtained by us would prove useful in organising the the kinematical contributions into smaller sets and also providing simplification in the computation by combining the matrices that diagonalise the mixing matrices with the kinematic contributions.

PUBLICATIONS

1. N. Agarwal, S. Pal, A. Srivastav and A. Tripathi, *CWebGen A tool to study colour structure of scattering amplitudes in IR limit*, *Comput. Phys. Commun.* **316** (2025) 109765 [2411.07872], doi: 10.1016/j.cpc.2025.109765
2. A. Danish, S. Mishra, S. Pal, A. Srivastav and A. Tripathi, *Color structure of next-to-eikonal correlator webs at three loops*, *Phys. Rev. D* **111** (2025) 114022 doi: 10.1103/xtft-b5dt
3. S. Mishra, S. Pal, A. Srivastav and A. Tripathi, *Multiparton Cwebs at five loops*, *JHEP* **07** (2024) 078 [2305.17452], doi: 10.1007/JHEP07(2024)078
4. N. Agarwal, S. Pal, A. Srivastav and A. Tripathi, *Correlator webs of massive multiparton amplitudes at four loops: A study of boomerang webs*, *Phys. Rev. D* **109** (2024) 094038 doi: 10.1103/PhysRevD.109.094038
5. N. Agarwal, M. van Beekveld, E. Laenen, S. Mishra, A. Mukhopadhyay and A. Tripathi, *Next-to-leading power corrections to event-shape variables*, *Pramana* **98** (2024) 60 doi: 10.1007/s12043-024-02743-0
6. M. van Beekveld, A. Danish, E. Laenen, S. Pal, A. Tripathi and C. D. White, *Next-to-soft radiation from a different angle*, *Phys. Rev. D* **109** (2024) 074005 [2308.12850], doi: 10.1103/PhysRevD.109.074005
7. N. Agarwal, S. Pal, A. Srivastav and A. Tripathi, *Deciphering colour building blocks of massive multiparton amplitudes at 4-loops and beyond*, *JHEP* **02** (2023) 258 [2212.06610], doi: 10.1007/JHEP02(2023)258
8. N. Agarwal, L. Magnea, C. Signorile-Signorile and A. Tripathi, *The infrared structure of perturbative gauge theories*, *Phys. Rept.* **994** (2023) 1 [2112.07099], doi: 10.1016/j.physrep.2022.10.001
9. N. Agarwal, S. Pal, A. Srivastav and A. Tripathi, *Building blocks of Cwebs in multiparton scattering amplitudes*, *JHEP* **06** (2022) 020 [2204.05936], doi: 10.1007/JHEP06(2022)020
10. N. Agarwal, L. Magnea, S. Pal and A. Tripathi, *Cwebs beyond three loops in multiparton amplitudes*, *JHEP* **03** (2021) 188 [2102.03598], doi: 10.1007/JHEP03(2021)188
11. N. Agarwal, A. Mukhopadhyay, S. Pal and A. Tripathi, *Power corrections to event shapes using eikonal dressed gluon exponentiation*, *JHEP* **03** (2021) 155 [2012.06842], doi: 10.1007/JHEP03(2021)155
12. N. Agarwal, A. Danish, L. Magnea, S. Pal and A. Tripathi, *Multiparton webs beyond three loops*, *JHEP* **05** (2020) 128 [2003.09714], doi: 10.1007/JHEP05(2020)128

13. N. Agarwal, P. Banerjee, G. Das, P. K. Dhani, A. Mukhopadhyay, V. Ravindran et al., *Resummed transverse momentum distribution of pseudo-scalar Higgs boson at NNLO_A+NNLL*, *JHEP* **12** (2018) 105 [[1805.12553](#)], doi: [10.1007/JHEP12\(2018\)105](#)
14. S. Alekhin et al., *A facility to Search for Hidden Particles at the CERN SPS: the SHiP physics case*, *Rept. Prog. Phys.* **79** (2016) 124201 [[1504.04855](#)], doi: [10.1088/0034-4885/79/12/124201](#)
15. R. V. Harlander, A. Tripathi and M. Wiesemann, *Higgs production in bottom quark annihilation: Transverse momentum distribution at NNLO+NNLL*, *Phys. Rev. D* **90** (2014) 015017 [[1403.7196](#)], doi: [10.1103/PhysRevD.90.015017](#)
16. N. Agarwal, V. Ravindran, V. K. Tiwari and A. Tripathi, *Next-to-leading order QCD corrections to W^+W^- production at the LHC in Randall Sundrum model*, *Phys. Lett. B* **690** (2010) 390 [[1003.5445](#)], doi: [10.1016/j.physletb.2010.05.063](#)
17. N. Agarwal, V. Ravindran, V. K. Tiwari and A. Tripathi, *W^+W^- production in Large extra dimension model at next-to-leading order in QCD at the LHC*, *Phys. Rev. D* **82** (2010) 036001 [[1003.5450](#)], doi: [10.1103/PhysRevD.82.036001](#)
18. N. Agarwal, V. Ravindran, V. K. Tiwari and A. Tripathi, *Next-to-leading order QCD corrections to the Z boson pair production at the LHC in Randall Sundrum model*, *Phys. Lett. B* **686** (2010) 244 [[0910.1551](#)], doi: [10.1016/j.physletb.2010.02.060](#)
19. N. Agarwal, V. Ravindran, V. K. Tiwari and A. Tripathi, *Z boson pair production at the LHC to $O(\alpha(s))$ in TeV scale gravity models*, *Nucl. Phys. B* **830** (2010) 248 [[0909.2651](#)], doi: [10.1016/j.nuclphysb.2009.12.032](#)
20. N. Agarwal, M. C. Kumar, P. Mathews, V. Ravindran and A. Tripathi, *Di-jet production at the LHC through unparticles*, *Phys. Rev. D* **80** (2009) 035015 [[0903.0202](#)], doi: [10.1103/PhysRevD.80.035015](#)
21. M. C. Kumar, P. Mathews, V. Ravindran and A. Tripathi, *Direct photon pair production at the LHC to order α_s in TeV scale gravity models*, *Nucl. Phys. B* **818** (2009) 28 [[0902.4894](#)], doi: [10.1016/j.nuclphysb.2009.03.022](#)
22. M. C. Kumar, P. Mathews, V. Ravindran and A. Tripathi, *Diphoton signals in theories with large extra dimensions to NLO QCD at hadron colliders*, *Phys. Lett. B* **672** (2009) 45 [[0811.1670](#)], doi: [10.1016/j.physletb.2009.01.002](#)
23. M. C. Kumar, P. Mathews, V. Ravindran and A. Tripathi, *Unparticles in diphoton production to NLO in QCD at the LHC*, *Phys. Rev. D* **79** (2009) 075012 [[0804.4054](#)], doi: [10.1103/PhysRevD.79.075012](#)
24. M. C. Kumar, P. Mathews, V. Ravindran and A. Tripathi, *Unparticle physics in diphoton production at the CERN LHC*, *Phys. Rev. D* **77** (2008) 055013 [[0709.2478](#)], doi: [10.1103/PhysRevD.77.055013](#)

PROCEEDINGS

Published

1. A. Danish, S. Mishra, S. Pal, A. Srivastav and A. Tripathi, *Next-to-eikonal Webs in Multiparton Amplitude*, *Springer Proc. Phys.* **304** (2024) 1114 doi: [10.1007/978-981-97-0289-3_307](https://doi.org/10.1007/978-981-97-0289-3_307), 25th DAE-BRNS High Energy Physics Symposium, Mohali, Punjab, Dec 2022.
2. A. Srivastav, N. Agarwal, S. Pal and A. Tripathi, *A Novel Approach to Understand the Color Structure of Multiparton Scattering Amplitudes*, *Springer Proc. Phys.* **304** (2024) 492 doi: [10.1007/978-981-97-0289-3_109](https://doi.org/10.1007/978-981-97-0289-3_109), 25th DAE-BRNS High Energy Physics Symposium, Mohali, Punjab, Dec 2022.
3. N. Agarwal, S. Pal, A. Srivastav and A. Tripathi, *Cwebs in multiparton scattering amplitude: Structures at four loops*, *EPJ Web Conf.* **314** (2024) 00024 doi: [10.1051/epjconf/202431400024](https://doi.org/10.1051/epjconf/202431400024), 11th International Workshop on QCD - Theory and Experiment (QCD@Work2024), Trani, Italy, June 2024.
4. S. Pal, N. Agarwal, A. Mukhopadhyay and A. Tripathi, *EDGE: A Simple Way to Obtain Power Corrections for Event Shape Variables*, *Springer Proc. Phys.* **304** (2024) 789 doi: [10.1007/978-981-97-0289-3_196](https://doi.org/10.1007/978-981-97-0289-3_196), 25th DAE-BRNS High Energy Physics Symposium, Mohali, Punjab, Dec 2022.
5. N. Agarwal, A. Danish, S. Mishra, S. Pal, A. Srivastav and A. Tripathi, *Soft Anomalous Dimension Structure at Five Loops*, *Springer Proc. Phys.* **304** (2024) 1117 doi: [10.1007/978-981-97-0289-3_308](https://doi.org/10.1007/978-981-97-0289-3_308), 25th DAE-BRNS High Energy Physics Symposium, Mohali, Punjab, Dec 2022.
6. N. Agarwal, M. van Beekveld, E. Laenen, S. Mishra, A. Mukhopadhyay and A. Tripathi, *Event shape distribution and the NLP corrections*, *EPJ Web Conf.* **314** (2024) 00023 doi: [10.1051/epjconf/202431400023](https://doi.org/10.1051/epjconf/202431400023), 11th International Workshop on QCD - Theory and Experiment (QCD@Work2024), Trani, Italy, June 2024.
7. S. Pal, N. Agarwal, L. Magnea and A. Tripathi, *Multiparton Webs Beyond Three Loops*, *Springer Proc. Phys.* **277** (2022) 253 doi: [10.1007/978-981-19-2354-8_46](https://doi.org/10.1007/978-981-19-2354-8_46), 24th DAE-BRNS High Energy Physics Symposium
8. N. Agarwal, A. Mukhopadhyay, S. Pal and A. Tripathi, *Eikonal Dressed Gluon Exponentiation to study power corrections to Thrust, C-parameter, and Angularity*, *SciPost Phys. Proc.* **7** (2022) 020 [2111.03365], doi: [10.21468/SciPostPhysProc.7.020](https://doi.org/10.21468/SciPostPhysProc.7.020), 15th International Symposium on Radiative Corrections: Applications of Quantum Field Theory to Phenomenology AND LoopFest XIX: Workshop on Radiative Corrections for the LHC and Future Colliders (Radcor and LoopFest 2021) 16-21 May 2021. Tallahassee, FL, United States (C21-05-16).

9. R. Harlander, A. Tripathi and M. Wiesemann, *Resummed Higgs p_T distribution at NNLO+NNLL in bottom-quark annihilation*, *PoS LL2014* (2014) 065 [[1407.3184](#)], doi: [10.22323/1.211.0065](#), 12th DESY Workshop on Elementary Particle Physics: Loops and Legs in Quantum Field Theory (LL2014), Weimar, Germany, 2014.
10. W^+W^- production at LHC at NLO in extra dimension models.
Neelima Agarwal, V. Ravindran, V. K. Tiwari, Anurag Tripathi.
Proceedings of Science (RADCOR 2011) 046.
ISSN 1824-8039, Co-Author
11. Working group report: Physics at the Large Hadron Collider.
D. K. Ghosh et al.
11th Workshop On High Energy Physics Phenomenology . Jan 2010, Ahmedabad, India.
Pramana J. Phys. 76 (2011) 691.
ISSN 0304-4289, Co-Author
12. Working group report: Quantum Chromodynamics sub-group.
Asmita Mukherjee et al.
The Tenth Workshop on High Energy Physics Phenomenology .
Jan 2008, Chennai, India
Pramana J. Phys. 72 (2009) 277.
ISSN 0304-4289, Co-Author
13. Diphoton production to NLO in QCD in the RS model at the LHC.
M. C. Kumar, Prakash Mathews, V. Ravindran, Anurag Tripathi.
Proceedings of workshop *Physics of Warped Extra Dimensions (PWED)*, Feb 2008, Kharagpur, India.
14. Unparticle physics v/s extra dimensions.
M. C. Kumar, Prakash Mathews, V. Ravindran, Anurag Tripathi.
Proceedings of workshop *Physics of Warped Extra Dimensions (PWED)*, Feb 2008, Kharagpur, India.
15. Working group report: Quantum Chromodynamics.
V. Ravindran et al.
The Ninth Workshop on High Energy Physics Phenomenology .
Jan 2008, Bhubaneswar, India
Pramana J. Phys. 67 (2006) 983.
ISSN 0304-4289, Co-Author

TALKS GIVEN AT CONFERENCES & WORKSHOPS

1. IMSc Spring School on High Energy Physics
24 Feb - 07 Mar 2025
IMSc, Chennai

Advanced topics in Quantum Field Theories
For recording [Click here](#)

2. CCAP 2025
30 Jan - 02 Feb, 2025
SRM, Chennai
Patterns in Scattering Amplitudes
3. AMPS 2024
28-30 Sep 2024
IISER Mohali, Mohali
Patterns in scattering amplitudes
4. India-CMS collaboration 2024 Lecture Series
01-05 July 2024, Online
Lectures on Infrared radiation and its cancellation in QED and QCD
Recording available at: <https://indiacms.res.in/>
5. Workshop in High Energy Physics Phenomenology (WHEPP XVII)
2-11 Jan 2024
IIT Gandhinagar
Deciphering the colour building blocks of multi-parton scattering amplitudes
6. International meeting on High Energy Physics (IMHEP -II)
16-22 Feb, 2023, IOP Bhubaneswar
Webs in Multiparton Scattering Amplitudes Beyond 3 loops
7. Resummation, Evolution, Factorization 2020
7-11 December 2020 (Online)
Multiparton Webs beyond three loops
8. GIAN course on Infrared Structure of Perturbative Gauge Theories
04-13 Dec 2017, IIT Hyderabad, India
Spinor-Helicity Methods in Gauge Theories
9. Saha Theory Workshop: Multi-loop and Multi-leg processes for precision physics at the LHC,
23-27 Feb 2016, Kolkata, India.
Generalized Unitarity to set up Feynman Master Integrals
10. Workshop on High Energy Physics Phenomenology,
04-13 Dec 2015, Kanpur, India.
Differential Equations Method and Generalized Unitarity for solving Feynman Integrals

11. International Workshop on Frontiers of QCD,
02-05 Dec 2014, Mumbai, India.
Log resummed distribution of the Higgs Bosons in bottom-quark fusion
12. International conference Loops and Legs in Quantum Field Theory,
27 April- 02 May 2014, Weimar, Germany.
Resummed Higgs p_T distribution at NNLL + NNLO in bottom fusion
13. 25th Rencontres de Blois on "Particle Physics and Cosmology",
May 26-31, 2013, Blois, France.
Power corrections to event shape distributions at e^+e^- colliders using eikonal approximation.
14. RADCOR 2011 Symposium. Radiative Corrections for the LHC: 2., Sep 26-30, 2011,
Mamallapuram, India.
 W^+W^- production at LHC at NLO in extra dimension models.
15. The 11th Workshop on High Energy Physics Phenomenology
(WHEPP XI), Jan 2 - 12, 2010, Ahmedabad, India.
Z boson pair production at the LHC at next-to-leading order in QCD in extra dimension models.
16. The Tenth Workshop on High Energy Physics Phenomenology
(WHEPP X), Jan 2 - 13, 2008, Chennai, India.
Diphoton production at the LHC at next-to-leading order in Unparticle scenario.
17. RECAP WORKSHOP 2008, Sep 21 - Oct 4, 2008, Allahabad, India.
Diphoton production at NLO in Unparticle scenario.
18. Signaling the Arrival of the LHC Era , Dec 08 - 13 , 2008, ICTP, Trieste, Italy.
Di-photon production at the LHC at next-to-leading-order in Unparticle scenario.
19. Topical meeting on Beyond the SM physics at the LHC,
Jan 15 - 17, 2009, Kolkata, India.
Photon pair production in ADD model at NLO at LHC.

VISITS AFTER 2015

- INFN Turin (06-14 May, 2024)
- INFN Milan (14-16 May, 2024)
- INFN Padova (16- 19 May, 2024)
- INFN Frascati (19-20 May, 2024)
- IIT BHU (20-23 March 2024)
- NIKHEF & University of Amsterdam (20 -23 June, 2018)

- University of Naples (18 - 20 June, 2018)
- University of Turin (19 May -18 June, 2018)
- IMSC Chennai
- IMSC Chennai

TEACHING EXPERIENCE

- Have been teaching an introductory course on Quantum Field Theory since 2015
- Taught Analytical Mechanics since 2016
- Taught Statistical Mechanics and Thermodynamics on and off since 2016.

REFERENCES

1. Prof. V. Ravindran (**Thesis Advisor**)
Institute of Mathematical Sciences
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3. Prof. Eric Laenen
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Member of CERN Council
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